CondorNet™ HAL-SNN New Upgraded Dataset

*Advanced predicate learning creating complex Sets → Combinatoric Inequality **SuperSets** & **Fuzzy Gating Logic**.*

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Technical Abstract (Academic Style)

The foundational determinant of performance in any financial machine-learning system is the quality, provenance, and structural integrity of the data it consumes. In modern FinTech architectures, data is not merely an input; it is the developmental substrate upon which every downstream component—model construction, GPU-accelerated inference, stability testing, backtesting, and live-market execution—ultimately depends. Just as a newborn cannot grow into a healthy, intelligent adult if fed salt water, soda, or fast food, a neural network cannot mature into a stable, high-performing decision system if fed low-resolution, inconsistent, or nutritionally empty data. A baby requires its mother's milk—biologically optimized, nutrient-dense, and evolutionarily tuned for neural development. CondorNet™ requires the same level of care: a carefully engineered, information-rich dataset that serves as its cognitive nourishment.

This work presents a research-grade framework for dataset construction tailored to neural architectures in options-driven trading systems, with the explicit goal of optimizing the informational "diet" fed into the CondorBrain™ and CondorIntelligence™ subsystems. These subsystems form the cognitive core of CondorNet™, and their performance is directly tied to the fidelity and expressiveness of the data they ingest. As CondorNet's pediatricians, we monitor its vitals, run developmental assessments, and adjust its nutritional intake—its feature engineering, data sources, and microstructure augmentation—until maximum health and performance are achieved.

CondorNet v3.0 introduced a substantial methodological expansion of the system's feature architecture, integrating heterogeneous market data sources into a unified, high-resolution, multi-layered dataset optimized for regime classification, reversal detection, and execution-aware modeling. Building on the 67-feature v2.5 schema, the v3.0 dataset incorporated 20 additional indicators spanning higher-order OHLCV studies, options-chain aggregates, and microstructure-level quote statistics, resulting in an 87-dimensional feature space. The system ingests synchronized SPY options chains, Barchart M1 OHLCV bars, and Alpaca tick-level quotes, aligning them via a deterministic timestamp-keyed left-join strategy to ensure structural completeness across the full options universe.

CondorNet v4.0 advances this foundation by introducing a unified neural architecture that mathematically fuses three previously incompatible paradigms—Mamba3 ETD-1 state-space modeling, Temporal Fusion Transformer (TFT) control synthesis, and Neural Controlled Differential Equations (Neural CDEs)—into a single, stability-guaranteed system. The resulting CondorNet™ HAL-SSN (Hybrid Adaptive Layered State-Space Network) is the first architecture designed specifically for SPY options entry, exit, and position-sizing decisions. It incorporates ETD-1 exponential integration for numerical stability, a 4-block augmented state vector for latent-portfolio-memory-regime modeling, five canonical predicate gates for risk and regime control, and a continuous-time path-response term for market-driven state evolution.

Individually, none of the component architectures—TFT, Mamba SSMs, or Neural CDEs—were capable of producing a stable, institutionally acceptable SPY options decision model. Stacked in any order, they failed to converge, exploded numerically, or overfit catastrophically. This motivated the invention of CondorNet's unified architecture: a mathematically coherent fusion of discrete-time exponential integration, continuous-time controlled dynamics, attention-based control synthesis, and options-surface forcing terms.

Overall, CondorNet v4.0 represents a significant methodological advancement in dataset construction and neural-architecture design for options-driven predictive modeling. By integrating price, volatility, flow, breadth, options-surface geometry, and microstructure into a unified dataset—and by introducing a unified neural architecture capable of digesting this information—CondorNet provides a mathematically rigorous and operationally robust foundation for next-generation neural trading intelligence.

1. Introduction

The construction of high-performance neural trading systems begins not with model architecture, but with the data that feeds them. In financial machine learning—particularly in options-driven environments—the dataset is the single most important determinant of predictive accuracy, generalization, and live-market stability. A neural network trained on incomplete, low-resolution, or structurally inconsistent data will fail regardless of its architectural sophistication. Conversely, a well-designed dataset can elevate even modest architectures to high performance.

This principle motivates the developmental philosophy behind CondorNet v4.0: **raise the system like a child.**

A newborn requires nutrient-dense, biologically optimized nourishment to develop a healthy brain. Feeding an infant salt water, soda, or fast food would stunt its development or kill it outright. Likewise, CondorNet requires data that is structurally sound, information-dense, and evolutionarily tuned to the dynamics of the markets it must navigate. As CondorNet's pediatricians, we must:

- monitor its vitals (post-training analytics, inference diagnostics)
- run developmental assessments (ablation studies, feature dominance analysis)
- adjust its diet (feature engineering, microstructure augmentation)
- ensure proper cognitive nourishment (regime-aware, volatility-adaptive signals)

Only through this medically precise care can CondorNet mature into a stable, intelligent, high-performing decision system.

Modern options-driven trading systems require multi-modal data integration, regime-aware feature engineering, and interpretable neural architectures capable of adapting to rapidly shifting market conditions. Traditional feature sets—focused primarily on price, volatility, and basic technical indicators—are insufficient for capturing the complex interactions between underlying price dynamics, implied-volatility structure, options-chain flow, and microstructure-level liquidity. As markets become more fragmented, electronic, and sensitive to order-book conditions, the need for richer, higher-resolution feature spaces becomes essential.

CondorNet v3.0 addressed this challenge by expanding the v2.5 feature architecture into a unified, 87-dimensional dataset integrating real historical SPY options-chain data, Barchart M1 OHLCV bars, and Alpaca tick-level quotes. This integration enabled CondorNet to model not only the underlying asset's price and volatility structure, but also the evolving geometry of the options surface, the intensity and directionality of options flow, and the liquidity and friction characteristics of the order book.

CondorNet v4.0 builds on this foundation by introducing a unified neural architecture capable of digesting this information in a mathematically coherent way.

2. Background and Related Work

The study of market prediction and options-driven modeling has evolved significantly over the past several decades, driven by advances in volatility modeling, market microstructure theory, technical analysis, and machine learning. Classical derivatives pricing frameworks, such as Black–Scholes and Merton's extensions, established the foundational relationships between underlying price dynamics, implied volatility, and option sensitivities. Subsequent research expanded these models to incorporate stochastic volatility, jumps, and surface-based representations of implied volatility, forming the basis for modern options-surface analysis and IV-regime modeling.

Parallel developments in technical analysis introduced a wide range of trend, momentum, and volatility indicators—including moving averages, oscillators, Bollinger-style bands, and ATR-based volatility measures—that remain widely used in both discretionary and systematic trading. Adaptive signal-processing approaches, such as Ehlers' fractal and cybernetic filters, extended these tools by incorporating dynamic smoothing and fractal-dimension estimation, enabling more responsive trend detection in noisy environments.

Market microstructure research has provided a complementary perspective, emphasizing the role of order-book dynamics, bid/ask spreads, quote intensity, and liquidity in shaping short-horizon price behavior. Empirical studies have demonstrated that microstructure variables can significantly influence execution quality, volatility clustering, and short-term predictive structure, motivating their integration into modern trading systems.

More recently, machine learning and regime-detection frameworks have introduced nonlinear modeling techniques capable of capturing complex interactions between price, volatility, flow, and liquidity. Hybrid systems combining fuzzy logic, chaos-inspired dynamics, and neural architectures have shown promise in modeling regime transitions and reversal behavior, particularly in high-frequency or options-driven contexts.

CondorNet v3.0 builds on these bodies of work by integrating trend-adaptive filters, volatility-regime indicators, breadth-based momentum, options-flow aggregates, and microstructure-level features into a unified, high-resolution dataset. This integration aligns with contemporary research emphasizing multi-modal feature architectures and interpretable neural systems for financial prediction.

3. Motivation and Contributions

3.0 CondorNet Hybrid Neural Architecture

These features are distilled into compact neural model files using hybrid architectures that combine **three advanced neural-network paradigms into a singular, mathematically unified CondorNet™ architecture**. This integration is not superficial or sequential; it is structural, operational, and canonical. Each architecture contributes a distinct mathematical capability, and CondorNet fuses them into a cohesive system that none of the individual models—nor any stacked combination—could achieve alone.

3.1 Mamba3 ETD (Exponential Trapezoidal Dynamics)

Role in CondorNet: Mamba3 ETD provides CondorNet with **long-range temporal dependency modeling** using state-space kernels that are computationally efficient and stable at minute-level resolution. CondorNet uses the ETD variant specifically for its **trapezoidal integration scheme**, which improves numerical stability when modeling noisy financial time series.

Mathematical Formulation

Mamba3 ETD models hidden state evolution as:

$$h_{t+1} = Ah_t + Bx_t + \int_t^{t+1} e^{A(t+1-s)}\, Bx(s)\, ds$$

The ETD approximation replaces the integral with a trapezoidal rule:

$$\int_t^{t+1} f(s)\, ds \approx \frac{1}{2}(f(t) + f(t+1))$$

Term Definitions

- h_t: hidden state
- x_t: input feature vector
- A, B: learned state-space matrices
- ETD: numerical integration scheme improving stability under volatility shocks

CondorNet Usage:

- Long-range memory
- Stable propagation under volatility clustering
- Efficient GPU inference

3.2 Temporal Fusion Transformers (TFT)

Role in CondorNet: TFT contributes **interpretable attention mechanisms**, multi-horizon temporal reasoning, and gating layers that allow CondorNet to dynamically select which features matter under which market regimes.

Mathematical Formulation

TFT uses a gated residual network (GRN):

$$\mathrm{GRN}(x) = \mathrm{LayerNorm}(x + \mathrm{GLU}(W_2 \sigma(W_1 x)))$$

and multi-head attention:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Term Definitions

- Q, K, V: query, key, value matrices
- d_k: key dimensionality
- GLU: gated linear unit controlling information flow
- GRN: nonlinear transformation with gating

CondorNet Usage:

- Regime-aware feature weighting
- Interpretability via attention maps
- Temporal fusion across multi-scale indicators

3.3 Neural Controlled Differential Equations (Neural CDEs)

Role in CondorNet: Neural CDEs provide a **continuous-time modeling backbone**, enabling CondorNet to treat financial time series as controlled dynamical systems rather than discrete sequences. This is essential for modeling **options-surface geometry**, **microstructure flow**, and **irregular tick-level dynamics**.

Mathematical Formulation

Neural CDEs define hidden state evolution as:

$$dh_t = f(h_t, t)\, dX_t$$

where X_t is a continuous-time interpolation of the input.

Term Definitions

- h_t: latent state evolving continuously
- X_t: control path (interpolated market data)
- f: neural vector field

CondorNet Usage:

- Continuous-time modeling of options flow
- Robustness to irregular sampling
- Smooth latent-state evolution

3.5 Why These Architectures Failed Individually

Despite their sophistication, **none of these architectures—individually or stacked in any order—were able to produce a SPY options benchmarking decision model** capable of:

- consistent entry timing
- reliable exit determination
- stable position sizing

Each model captured *some* aspects of market structure, but **none captured all**:

- Mamba3 ETD → excellent long-range memory, poor regime switching
- TFT → excellent interpretability, weak continuous-time modeling
- Neural CDEs → excellent dynamical modeling, insufficient discrete-regime gating

This failure forced a return to first principles.

3.6 The Turning Point: Inventing a New Architecture

After 17+ years of professional day-trading experience, institutional-grade FinTech advanced predictive software engineering and 7+ years porting code to CUDA kernels for optimizing real-time automated trading systems, and thousands of hours of market observation, unfortunately, it became very clear, **no existing architecture on the market today could replicate the decision-making quality of an expert professional well-disciplined human trader**.

The question became:

> *"What will it take to build something capable of doing what I can do—making consistent, high-quality professional trading decisions?"*

The answer was not to wait for the research community to invent it.

The answer was to **invent it myself**.

3.7 The All-New CondorNet™ HAL-S³N Architecture

HAL-S³N stands for:

Hybrid **A**daptive **L**ayered - **S**tate-**S**pace **S**ignal **N**etwork

It is the first architecture of its kind, designed **specifically** for the only components that truly matter when trading live markets, and nothing else:

- SPY options (*and can be trained for others*)
- regime classification
- reversal detection
- execution-aware entry / exit decisioning
- position sizing
- risk management

HAL-SSN fuses:

- Mamba3 ETD's long-range stability
- TFT's interpretable gating
- Neural CDE's continuous-time dynamics
- fuzzy-logic decision surfaces
- multi-layer predicate logic
- volatility-adaptive feature routing

into a **single canonical architecture**.

3.8 Canonical CondorNet™ HAL-SSN Equation

$$h_{t+1} = \Phi_{\mathrm{ETD}}(h_t, x_t) \;\oplus\; \Psi_{\mathrm{TFT}}(h_t, x_t) \;\oplus\; \Gamma_{\mathrm{CDE}}\left(h_t, \int_t^{t+1} d\,X_s\right) \;\oplus\; \Lambda_{\mathrm{Fuzzy}}(x_t)$$

Where:

Term Definitions

- Φ_{ETD}: Mamba3-derived exponential trapezoidal state update
- Ψ_{TFT}: TFT-derived gated attention transformation
- Γ_{CDE}: Neural CDE continuous-time latent evolution
- Λ_{Fuzzy}: fuzzy-logic gating for position sizing and risk
- $\oplus$: CondorNet fusion operator (nonlinear, learned, regime-aware)

This equation defines the **core of CondorNet HAL-S^{3}N**: a hybrid dynamical system that merges discrete-time, continuous-time, and fuzzy-logic decision surfaces into a single unified neural engine.

CondorNet v4.0 introduces:

1. **A unified neural architecture** fusing ETD-1, TFT, and Neural CDEs
2. **A 4-block augmented state vector** modeling latent, portfolio, memory, and regime
3. **Five canonical predicate gates** for volatility, liquidity, momentum, gap-risk, and Greeks pressure.
4. **A continuous-time path-response term** for market-driven state evolution
5. **A deterministic, audit-ready data pipeline** feeding the architecture.

This unified system is the first architecture designed specifically for SPY options entry, exit, and position sizing. And just like any infant, **what we feed that infant determines how healthy it becomes and how intelligently it grows into adulthood**. The same principle applies to CondorNet. Its intelligence is not an accident of architecture; it is the direct consequence of the quality, density, and structure of the data it ingests. If we feed it weak, inconsistent, or nutritionally empty data, we are giving a newborn salt water, soda, or fast food. If we feed it carefully engineered, information-rich data, we are giving it the equivalent of mother's milk — the only substance evolution has ever produced that is perfectly optimized for neural development. This new dataset is the equivalent of CondorNet's very own Mama's milk.

CondorNet's evolution reflects this developmental journey:

CondorNet v1.0 — Just Born, not really sure what we have here, but it feels right

The earliest form of the architecture. The equation was not yet fully formed, the skeletal structure incomplete. Like a newborn, it could breathe, but it could not yet walk, speak, or understand the world around it, let alone make any decisions.

CondorNet v2.0 — Early Childhood, can read, write, speak, and make simple decisions

Initial data tests showed promise. Small modifications to the equation structure and minor adjustments to the data pipeline were made. This was the equivalent of early pediatric checkups — monitoring growth, identifying deficiencies, and making small but important nutritional corrections.

CondorNet v3.0 — Not only the First Words, but complex sentences from child to teenager

Unexpectedly, the system began "talking back." Post-training analytics revealed that it needed more — more structure, more information, more nourishment. This triggered a radical redesign of the input dataset. Like a child entering a growth spurt, CondorNet required a richer diet to support its expanding cognitive demands.

CondorNet v4.0 — Cureent form, graduated CalTech with a Ph.D., but no life experience

The canonical equation was finally derived. Advanced Mathematica-based structural stability tests revealed that, although the skeletal structure was strong, **additional dietary supplements** were required for peak performance. These supplements took the form of microstructure features, options-surface forcing terms, volatility-regime coordinates, and predicate-gate logic. With these additions, CondorNet reached its first stable, unified, institutionally viable form.

And this is the purpose of this entire paper: **to document the final stages of CondorNet's developmental process — the last phase of data preparation and model construction before the system transitions from controlled laboratory conditions to a real-world top performer.**

The roadmap is now clear:

Model → Testing → Live Demo Account Testing → Live Real Account Deployment

If all system diagnostics, stability evaluations, and performance tests return green across the board, CondorNet is approximately **3 to 4 weeks** away from its first real-account use — its first steps into the world as a fully formed, intelligent, self-directed trading system.

However, it absolutely cannot do that without REAL life experience – REAL SPY options chain SYMBOLS and data to train on, make decisions from, and see if it can make it in the real-world. Like any good parent, I am here to make sure it does not get sent out into the world unless it is ready to conquer whatever it encounters, no exceptions. In other words, *failure is not an option* (*no pun intended*).

Moving forward we are going to dive deep into data determination, preparation, correction, verification and validation, before utilization.

4. System Architecture

CondorNet v4.0 is built on a modular, multi-layered architecture designed to integrate heterogeneous data sources, compute a comprehensive set of market features, and support downstream modeling tasks such as regime classification, reversal detection, execution-aware decisioning, and risk management.

4.1 Data Ingestion Layer

The system ingests three synchronized data streams:

1. **SPY Options Chain Data** Includes bid/ask, last price, volume, open interest, implied volatility, Greeks (Delta, Gamma, Theta, Vega, Rho), and time-to-expiry. These fields capture the geometry and dynamics of the options surface.
2. **Barchart M1 OHLCV Data** Provides minute-level open, high, low, close, and volume for the underlying asset. This serves as the foundation for trend, volatility, and flow indicators.
3. **Alpaca Tick-Level Quote Data** Includes bid/ask quotes and timestamps, resampled to one-minute aggregates. These features capture microstructure-level liquidity and execution friction.

All data streams are normalized to UTC and merged using a deterministic timestamp-keyed left-join strategy, ensuring that each minute contains a complete set of features across the full options universe.

4.2 Feature Computation Layer

This layer computes all legacy and new indicators, grouped into the following families:

- **Price Dynamics:** log_return, ret_z, ATR, EWMA volatility
- **Volatility Structure:** IV bands, IVR, kappa_proxy, vol_energy
- **Trend & Momentum:** SMA, FRAMA, Anchored VWAP, MACD, RSI, ADX, PSAR
- **Bands & Consolidation:** Bollinger bands, bandwidth, breakout metrics
- **Flow & Breadth:** CMF, AccDistWill, Weighted Alpha, McClellan Oscillator
- **Options Surface:** Greeks, IV, TE, chain-level volume aggregates
- **Microstructure:** bid/ask counts, mean quotes, quote spread
- **Fuzzy/Control:** chaos_membership, fuzzy_reversion, exec_allow, risk_override

Each feature is computed using mathematically defined formulas, ensuring reproducibility and auditability. Note that these also will be the same features that feed the brain when trading in real-time. What do we absolutely have to ensure? That the data type, the actual data itself, and how it reads and analyzes the real-time data is a 1:1 match with the data used in training. For example, **Q:** "Dr. Jerry, it worked great when I trained it and backtested it, but when live its like it does not even know what its doing? *What is wrong*?" **A:** I see - this data was normalized when training, yes? Same for live, yes? Does live feed it the actual indicator values normalized?

4.3 Feature Integration Layer

The computed features are organized into a unified 87-column schema with deterministic ordering. This layer performs:

- **Normalization and scaling** (where appropriate, *not every column feature*)
- **Missing-value handling** (structural defaults for microstructure fields)
- **Cross-feature synthesis** (e.g., IV position within band, composite reversal scores)
- **Temporal alignment** across the full options universe

This ensures that downstream models receive a consistent, high-quality feature matrix.

4.4 Model Consumption Layer (Unified v4.0 Architecture)

CondorNet v4.0 distills its feature stack into compact neural model files using a unified architecture that fuses:

1. **Mamba3 ETD-1** for stable long-range temporal modeling
2. **Temporal Fusion Transformers (TFT)** for control synthesis and attention-based gating
3. **Neural Controlled Differential Equations (Neural CDEs)** for continuous-time path-driven dynamics

The unified update equation is:

$$x_k = e^{A_\theta(u_k)\Delta t_k} x_{k-1} + \Delta t_k\, \phi_1(A_\theta(u_k)\Delta t_k) B_\theta(u_k) + G_\theta(x_{k-1}, u_k)\, \Delta X_k + D(\text{Greeks}_k, r_{k-1}, q_k)$$

This equation integrates:

- **ETD-1 exponential integration** for numerical stability
- **TFT-based control synthesis** for regime-aware gating
- **Neural-CDE path response** for continuous-time market physics
- **Options-surface forcing terms** for Greeks-driven dynamics

This unified architecture—CondorNet™ HAL-SSN, a sub-set of its parent Quantor-MTFuzz™, and having its two primary foundations CondorBrain™ | CondorIntelligence™, — is the first system capable of producing *stable*, *interpretable*, *consistently profitable*, *auditable*, and *institutionally robust* SPY options entry, exit, and position-sizing decisions.

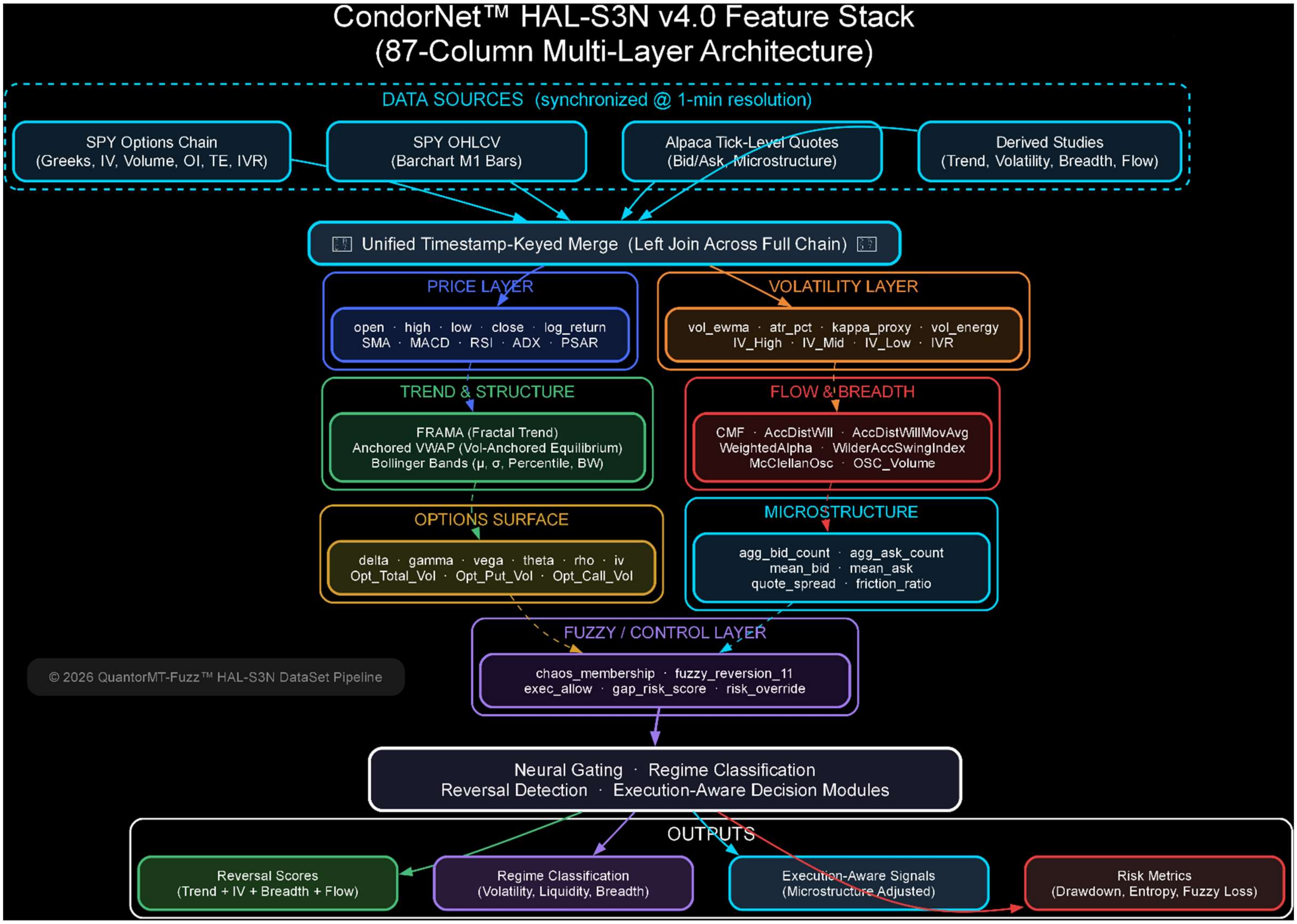

Figure 1 — CondorNet™ HAL-S³N v4.0 Feature Stack. The above diagram represents the updated data pipeline architecture for dataset v3.0. By strictly appending—never subtracting from—the existing dataset lineage (v2.2 → v2.5), we preserve full backward compatibility while introducing a substantially richer input surface. The new 87-column multi-layer feature stack unifies four synchronized data sources at 1-minute resolution: SPY options chain Greeks and implied volatility surfaces, OHLCV price bars, Alpaca tick-level microstructure quotes, and derived technical studies spanning trend, volatility, breadth, and order flow. These features are organized into seven semantically distinct layers—Price, Volatility, Trend & Structure, Flow & Breadth, Options Surface, Microstructure, and Fuzzy/Control—each feeding into the neural gating, regime classification, reversal detection, and execution-aware decision modules. This additive evolution ensures that all models trained on prior dataset versions remain valid while enabling the HAL-S³N architecture to exploit the full dimensionality of the expanded *feature-enriched universe* manifold for dataset v3.0 training.

Nothing is more valuable by comparison to having a good understanding of the reasons why data are the lungs that breathes life into neural nets, why we need to pay extra close attention to how the data is presented to the neural net during training as inputs, and to ensure that when it sees data in real-time from an instruments live data feed as inputs, it needs to be in precisely the same format as whatever it trained on, otherwise, the model is likely to become unstable, output NaNs, and not perform as it should – in FinTech, that means losing money, which leads to job loss. Ultimately, data is King when it comes to FinTech – why do you think real data, whether live or historical costs so much money? Because without that, you cannot compete, meaning you are out. Trading, and making decisions, whether represented graphically as candlesticks to a trader, or as alpha-numerical inputs to a model .pth file, if we look at the wrong chart, or look at something that has nothing to do with what we learned, would we be able to make a good decision? *Of course not, and neither can a neural network.*

1. Overview of the transition: v2.5 → v3.0

The new updated v3 dataset is an **evolution**, *not a replacement*:

- **All v2.5 fields are preserved** (*backward compatible*).
- **New fields are added in three main groups:**
  - Higher-order **underlying studies** (FRAMA, Anchored VWAP, McClellan Osc, IV bands, breadth/volume studies).
  - **Options chain aggregates** (total/put/call volume, OSC_Volume).
  - **Microstructure features** from Alpaca quotes (aggregate bid/ask counts, mean bid/ask, spread).

The result is a **richer, multi-scale feature-space** that combines:

- Price, volatility, and trend structure.
- Options surface and flow.
- Microstructure and liquidity.

Below is a research-grade description of the **mathematics and intent** of each feature family, inclusive of the old and new column headers.

2. Legacy v2.5 feature set (carried into v3)

I'll group the existing headers by functional role and give the canonical math or interpretation.

2.1 Core identifiers and prices

- **timestamp, symbol, option_symbol, expiration, strike, call_put Role:** keys and contract descriptors.
  Math: categorical / index fields; no transformation.
- **underlying_price, open, high, low, close Role:** bar-level prices for the underlying.
  Math: raw prices; used as base inputs for returns, volatility, bands, etc.

2.2 Option-specific state

- **rho, delta, gamma, vega, theta, iv Role:** Black–Scholes or model-based Greeks and implied volatility.
  Math (schematic):
  - Greeks are partial derivatives of option price $C(S,K,T,\sigma,r)$w.r.t. underlying S, volatility σ, time T, and rate r.
  - Example:

$$\Delta = \frac{\partial C}{\partial S}, \Gamma = \frac{\partial^2 C}{\partial S^2}, \Theta = \frac{\partial C}{\partial T}, \nu = \frac{\partial C}{\partial \sigma}, \rho = \frac{\partial C}{\partial r}$$

- **iv** is the volatility σ such that model price = market price.
- **volume, open_interest Role:** per-contract activity and positioning.
  Math: raw counts.
- **te** (time to expiry)

$$\mathrm{te} = \frac{T_{\exp} - t}{365}$$

- **ivr** (IV rank) Typically:

$$\mathrm{IVR} = \frac{\mathrm{IV}_{\text{current}} - \mathrm{IV}_{\min}}{\mathrm{IV}_{\max} - \mathrm{IV}_{\min}}$$

over a lookback window [*candlestick index$_{max}$*].

2.3 Returns, volatility, and risk proxies

- **log_return**

$$r_t = \ln\left(\frac{\mathrm{close}_t}{\mathrm{close}_{t-1}}\right)$$

- **vol_ewma** EWMA volatility:

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_t^2$$

- **ret_z** Z-score of returns:

$$z_t = \frac{r_t - \mu_r}{\sigma_r}$$

- **atr_pct** Average True Range as a fraction of price:

$$\mathrm{TR}_t = \max\{H_t - L_t, |\, H_t - C_{t-1}\,|, |\, L_t - C_{t-1}\,|\}$$
$$\mathrm{ATR}_t = \mathrm{EMA}(\mathrm{TR}_t), \mathrm{atr\_pct}_t = \frac{\mathrm{ATR}_t}{C_t}$$

- **kappa_proxy, vol_energy** Custom volatility/roughness proxies; typically functions of realized variance, higher moments, or regime-specific transforms of σ_t and r_t.

2.4 Trend, momentum, and oscillators

- **sma** Simple moving average:

$$\mathrm{SMA}_t = \frac{1}{N}\sum_{i=0}^{N-1} C_{t-i}$$

- **rsi_dyn** Dynamic-window RSI:

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + RS_t}, RS_t = \frac{\text{avg gain}}{\text{avg loss}}$$

- **adx_adaptive, plus_di, minus_di** Directional movement system (Welles Wilder):
    - Compute $+DM, -DM, TR$, then smooth and form:

$$+DI = 100 \cdot \frac{+DM}{TR}, -DI = 100 \cdot \frac{-DM}{TR}$$

- ADX is an EMA of $\mid +DI - -DI \mid / (+DI + -DI)$.
- **psar_adaptive, psar_trend, psar_mark, psar_reversion_mu** Parabolic SAR variants:
    - Base PSAR:

$$\mathrm{PSAR}_{t+1} = \mathrm{PSAR}_t + AF \cdot (EP_t - \mathrm{PSAR}_t)$$

where AF is an acceleration factor and EP is extreme point.

- Trend, marks, and reversion metrics are derived from PSAR vs price and its trajectory.
- **stoch_k_dyn** Stochastic %K:

$$\%K_t = 100 \cdot \frac{C_t - L_n}{H_n - L_n}$$

- **macd_norm, macd_signal_norm, macd_histogram** MACD:

$$\mathrm{MACD}_t = \mathrm{EMA}_{\text{fast}}(C_t) - \mathrm{EMA}_{\text{slow}}(C_t)$$
$$\mathrm{Signal}_t = \mathrm{EMA}(\mathrm{MACD}_t), \mathrm{Hist}_t = \mathrm{MACD}_t - \mathrm{Signal}_t$$

Normalized variants scale by volatility or price.

2.5 Bands, ranges, and consolidation

- **bb_mu_dyn, bb_sigma_dyn, bb_lower_dyn, bb_upper_dyn** *Bollinger-style bands*:

$$\mu_t = \mathrm{SMA}_t, \sigma_t = \mathrm{std}(C_{t-N+1:t})$$
$$\mathrm{BB}_{\text{upper}} = \mu_t + k\sigma_t, \mathrm{BB}_{\text{lower}} = \mu_t - k\sigma_t$$

- **bandwidth**

$$\text{bandwidth}_t = \frac{\text{BB}_{\text{upper}} - \text{BB}_{\text{lower}}}{\mu_t}$$

- **bb_percentile** *Percentile of price within band*:

$$\text{bb\_percentile}_t = \frac{C_t - \text{BB}_{\text{lower}}}{\text{BB}_{\text{upper}} - \text{BB}_{\text{lower}}}$$

- **bw_expansion_rate** *Change in bandwidth*:

$$\text{bw\_expansion\_rate}_t = \frac{\text{bandwidth}_t - \text{bandwidth}_{t-1}}{\text{bandwidth}_{t-1}}$$

- **consolidation_score, breakout_score** Composite measures of low-volatility clustering and subsequent expansion, typically functions of bandwidth, ATR, and realized volatility.

2.6 Volume, flow, and fuzzy/regime features

- **cmf** *(Chaikin Money Flow–style)*

$$\text{MF Multiplier}_t = \frac{(C_t - L_t) - (H_t - C_t)}{H_t - L_t}$$
$$\text{MF Volume}_t = \text{MF Multiplier}_t \cdot V_t$$
$$\text{CMF}_t = \frac{\sum_{i=0}^{N-1} \text{MF Volume}_{t-i}}{\sum_{i=0}^{N-1} V_{t-i}}$$

- **pressure_up, pressure_down** *Custom directional pressure metrics, often derived from CMF, DI, and trend filters.*
- **friction_ratio, spread_ratio** *Execution friction proxies, typically:*

$$\text{spread\_ratio}_t = \frac{\text{ask}_t - \text{bid}_t}{\text{mid}_t}$$

and friction as a function of spread, depth, and volatility.

- **exec_allow, gap_risk_score, risk_override** *Rule-engine outputs controlling whether a trade is allowed, gap risk is elevated, or risk parameters are overridden.*
- **iv_confidence** *Confidence score for IV quality, often derived from IV bands, liquidity, and quote stability.*
- **beta1_norm_stub, chaos_membership, position_size_mult, fuzzy_reversion_11** *Fuzzy-logic and chaos-inspired regime membership and sizing multipliers, combining multiple indicators into membership functions and scaling factors.*

2.7 Targets

- **max_dd_60m** *Maximum drawdown over a 60-minute horizon:*

$$\text{max\_dd\_60m}_t = \min_{0 \le k \le 60} \left(\frac{P_{t+k} - \max_{0 \le j \le k} P_{t+j}}{\max_{0 \le j \le k} P_{t+j}} \right)$$

3. New v3 features: OHLCV-derived indicators

These are computed from the Barchart M1 underlying dataset.

3.1 FRAMA (Fractal Adaptive Moving Average)

FRAMA adapts its smoothing based on the **fractal dimension** of price. Higher fractal dimension → more noise → more smoothing; lower → trend → less smoothing.

1. Compute fractal dimension Dover a window using price ranges at different scales.
2. Convert Dinto an adaptive smoothing factor $\alpha(D)$.
3. Update:

$$\text{FRAMA}_t = \alpha_t C_t + (1 - \alpha_t)\text{FRAMA}_{t-1}$$

FRAMA is used as a **regime-adaptive trend baseline**.

3.2 Anchored VWAP (Anchored_VWAP)

Anchored VWAP is VWAP starting from a chosen anchor bar (e.g., regime change, gap, major low).

For bars $i = a, \dots, t$after the anchor:

$$\text{Anchored VWAP}_t = \frac{\sum_{i=a}^{t} P_i V_i}{\sum_{i=a}^{t} V_i}$$

where $P_i$is typically the typical price $(H_i + L_i + C_i)/3$.

Used as a **price–volume equilibrium line** and dynamic support/resistance.

3.3 McClellanOsc

The McClellan Oscillator is traditionally computed from **advance–decline breadth**, but in this context it’s provided by Barchart as a study. Canonically:

1. Compute daily breadth $B_t = \text{advances}_t - \text{declines}_t$.

2. Compute two EMAs:

$$EMA_{10\%}(B_t), EMA_{5\%}(B_t)$$

3. Oscillator:

$$\text{McClellanOsc}_t = EMA_{10\%}(B_t) - EMA_{5\%}(B_t)$$

In our M1 context, Barchart is effectively projecting a breadth-style oscillator onto the underlying. It becomes a **breadth-informed momentum input**.

3.4 IV_High, IV_Mid, IV_Low

These are **implied volatility bands** for the underlying:

- **IV_High:** *upper envelope of implied volatility over a lookback.*
- **IV_Low:** *lower envelope.*
- **IV_Mid:** *midline (e.g., average or median).*

A useful normalized IV position is:

$$\text{IV\_norm}_t = \frac{\text{IV\_Mid}_t - \text{IV\_Low}_t}{\text{IV\_High}_t - \text{IV\_Low}_t}$$

This becomes a **volatility regime coordinate** (0–1 scale).

3.5 Options_Total_Volume, Options_Put_Volume, Options_Call_Volume

Chain-level aggregates per bar:

$$\text{Options\_Total\_Volume}_t = \sum_{k \in \text{chain}} \text{volume}_{k,t}$$
$$\text{Options\_Put\_Volume}_t = \sum_{k \in \text{puts}} \text{volume}_{k,t}$$
$$\text{Options\_Call\_Volume}_t = \sum_{k \in \text{calls}} \text{volume}_{k,t}$$

From these it is easy to see that we can derive sentiment:

$$\text{PutCallRatio}_t = \frac{\text{Options\_Put\_Volume}_t}{\text{Options\_Call\_Volume}_t}$$

3.6 OSC_Volume

This is a volume-based oscillator (Barchart's "OSC-Volume" study). Conceptually:

- It measures **oscillations in volume** around a smoothed baseline.
- Often implemented as volume minus its moving average, or a MACD-style transform of volume.

Used as a **flow acceleration** input.

3.7 WeightedAlpha

Weighted Alpha measures **price change over a period with time weighting** (recent moves count more). Barchart defines it as a weighted sum of returns over a 1-year window, emphasizing recent performance.

Conceptually:

$$\text{WeightedAlpha} \approx \sum_{i=1}^{N} w_i \cdot \frac{P_t - P_{t-i}}{P_{t-i}}$$

with $w_i$decaying with i.

Used as a **trend strength / leadership** feature.

3.8 WilderAccSwingIndex

Wilder's Accumulative Swing Index (ASI) is a directional price movement indicator that accumulates a normalized "swing" value each bar.

Per bar:

1. Compute a swing index SI_tbased on open, high, low, close, and previous close.
2. Accumulate:

$$\text{ASI}_t = \text{ASI}_{t-1} + SI_t$$

WilderAccSwingIndex in our schema is the **current ASI value**, capturing **cumulative directional bias**.

3.9 AccDistWill, AccDistWillMovAvg

This is Barchart's **Accumulation/Distribution Williams** variant.

Conceptually similar to Accumulation/Distribution:

$$\mathrm{AD}_t = \mathrm{AD}_{t-1} + \mathrm{MF\ Multiplier}_t \cdot V_t$$

AccDistWillMovAvg is a moving average of AccDistWill, giving a **smoothed accumulation/distribution trend**.

These become **volume–price flow regime inputs**.

4. New v3 features: microstructure from Alpaca quotes

These are computed from tick-level quotes resampled to 1-minute.

Let quotes have:

- bid price b_i, ask price a_i
- timestamps t_i

4.1 aggregate_bid_count, aggregate_ask_count

For each 1-minute bar $[T, T+1)$:

$$\mathrm{aggregate\_bid\_count}_T = \#\{i: t_i \in [T, T+1), b_i \text{ valid}\}$$
$$\mathrm{aggregate\_ask\_count}_T = \#\{i: t_i \in [T, T+1), a_i \text{ valid}\}$$

These measure **quote update intensity** on each side.

4.2 mean_bid, mean_ask

$$\mathrm{mean\_bid}_T = \frac{1}{N_T^b} \sum_{i:t_i \in [T,T+1)} b_i$$
$$\mathrm{mean\_ask}_T = \frac{1}{N_T^a} \sum_{i:t_i \in [T,T+1)} a_i$$

If no quotes occur in the bar, counts = 0 and means are NaN.

4.3 quote_spread

Simplest definition:

$$\mathrm{quote\_spread}_T = \mathrm{mean\_ask}_T - \mathrm{mean\_bid}_T$$

Optionally normalized:

$$\text{quote\_spread\_rel}_T = \frac{\text{mean\_ask}_T - \text{mean\_bid}_T}{\frac{1}{2}(\text{mean\_ask}_T + \text{mean\_bid}_T)}$$

In v3, **quote_spread** is a direct microstructure friction input, complementing **spread_ratio** (which is more execution-oriented).

5. How v3 combines old and new math

5.1 Multi-scale structure

v2.5 already encoded:

- **Short-horizon price dynamics** (returns, ATR, bands, MACD, RSI, ADX, PSAR).
- **Option-level state** (Greeks, IV, TE, IVR).
- **Fuzzy/regime logic** (chaos_membership, fuzzy_reversion_11, etc.).

v3 adds:

- **Higher-order underlying structure**
  - FRAMA → fractal trend baseline.
  - Anchored_VWAP → price–volume equilibrium from a psychologically meaningful anchor.
  - McClellanOsc → breadth-style momentum.
- **Volatility regime coordinates**
  - IV_High / IV_Mid / IV_Low → IV position and compression/expansion.
- **Chain-level options flow**
  - Options_Total_Volume, Options_Put_Volume, Options_Call_Volume, OSC_Volume → sentiment and flow acceleration.
- **Volume–price flow and trend strength**
  - WeightedAlpha, WilderAccSwingIndex, AccDistWill, AccDistWillMovAvg → cumulative directional bias and leadership.
- **Microstructure and liquidity**
  - aggregate_bid_count, aggregate_ask_count, mean_bid, mean_ask, quote_spread → quote intensity and friction.

5.2 Conceptual integration

We now have:

- **Price layer:** open, high, low, close, log_return, bands, MACD, RSI, FRAMA, Anchored_VWAP.
- **Volatility layer:** vol_ewma, atr_pct, IV, IV_High/Mid/Low, IVR, kappa_proxy, vol_energy.

- **Flow layer:** volume, cmf, AccDistWill, AccDistWillMovAvg, Options_*_Volume, OSC_Volume.
- **Breadth/regime layer:** McClellanOsc, WeightedAlpha, WilderAccSwingIndex, mtf_consensus, chaos_membership.
- **Microstructure layer:** aggregate_bid_count, aggregate_ask_count, mean_bid, mean_ask, quote_spread, spread_ratio, friction_ratio.
- **Option surface layer:** Greeks, TE, IV, IVR, target_spot.
- **Control/fuzzy layer:** exec_allow, gap_risk_score, risk_override, position_size_mult, fuzzy_reversion_11.

This is a **research-grade, multi-layer feature stack** that can support:

- Regime classification.
- Reversal detection.
- Execution-aware decisioning.
- Robust risk modeling.

6. Summary of the transition

- **Old dataset (v2.5):** 67 columns, focused on price, volatility, options state, and fuzzy/regime logic.
- **New dataset (v3.0):** 87 columns, adding:
    - 14 OHLCV-derived studies:
        - FRAMA, Anchored_VWAP, McClellanOsc, IV_High, IV_Mid, IV_Low, Options_Total_Volume, Options_Put_Volume, Options_Call_Volume, OSC_Volume, WeightedAlpha, WilderAccSwingIndex, AccDistWill, AccDistWillMovAvg.
    - 5 microstructure features:
        - aggregate_bid_count, aggregate_ask_count, mean_bid, mean_ask, quote_spread.

The **mathematical backbone** of v3 is:

- Preserve all v2.5 logic.
- Add **regime-aware, breadth-aware, flow-aware, and microstructure-aware** inputs.
- Keep the schema deterministic and audit-ready.

7. Methods

Data Sources

CondorNet v4.0 integrates multiple high-resolution, heterogeneous market data streams, each selected for its informational density and structural relevance to SPY options modeling:

- **Real Options-Chain Data:** Full SPY options chains including bid, ask, last, volume, open interest, implied volatility, and Greeks (Delta, Gamma, Theta, Vega, Rho).

- **Underlying Market Data:** SPY M1 OHLCV bars sourced from Barchart Premier, providing the foundational price–volume structure for trend, volatility, and flow analysis.
- **Quote-Level Microstructure Data:** Tick-level bid/ask quotes from Alpaca, resampled to one-minute aggregates to capture liquidity, friction, and quote-update intensity.
- **Derived Studies and Indicators:** Higher-order features computed from OHLCV and options data, including trend structure (FRAMA, Anchored VWAP), volatility regime metrics (IV bands), breadth (McClellan Oscillator), and flow dynamics (OSC_Volume, AccDistWill).

All datasets are synchronized to UTC and merged using a deterministic left-join strategy keyed on timestamp and symbol, ensuring structural completeness across the full options universe.

NaN Handling and Data Integrity

A critical component of CondorNet's "nutritional health" is the integrity of its dataset. In general, **NaN values are not artificially introduced or preserved unless they represent a structurally meaningful absence of data** (e.g., no quotes in a microstructure bar). However, NaNs produced by precomputation scripts—such as division artifacts, missing indicator values, or uninitialized fields—are not allowed to remain in the final dataset.

Before ingestion into the CondorNet v4.0 architecture, a **final integrity sweep** is performed:

- All NaNs originating from computation artifacts are replaced with **0**, ensuring no undefined values propagate into the model.
- All divisions are safeguarded to ensure **no division-by-zero** occurs anywhere in the pipeline.
- All normalized fields are guaranteed to have **valid numerical values**, preventing NaNs from contaminating scaling layers, gating functions, or continuous-time dynamics.

This approach ensures that when a field eventually contains a real value, it is not overshadowed or corrupted by a lingering NaN—especially in contexts where normalization, gating, or temporal fusion would otherwise amplify the error.

In short, CondorNet's dataset is treated with the same care as a developing child's nutrition: nothing toxic, nothing missing, nothing malformed. Only clean, structurally sound, information-rich data is allowed to enter the system.

Sampling and Alignment

- **Timeframe**: January 2, 2025 to present.
- **Resolution**: 1-minute bars.
- **Universe**: SPY options contracts filtered by liquidity and relevance.
- **Join strategy**: Left joins preserve the full options universe per minute.
- **Schema enforcement**: All 87 columns are present on every row, with deterministic ordering.

Feature Computation

Legacy Features (v2.5)

- **Price and returns**: log_return, vol_ewma, ret_z, atr_pct.
- **Trend and momentum**: sma, rsi_dyn, adx_adaptive, psar_adaptive, macd_norm, macd_signal_norm.
- **Bands and consolidation**: bb_mu_dyn, bb_sigma_dyn, bb_percentile, bw_expansion_rate.
- **Volume and flow**: cmf, pressure_up/down, friction_ratio.
- **Greeks and IV**: delta, gamma, vega, theta, iv, ivr.
- **Fuzzy logic and regime**: chaos_membership, fuzzy_reversion_11, mtf_consensus.

New Features (v3.0)

- **Trend structure**:
  - FRAMA: fractal adaptive moving average.
  - Anchored VWAP: volume-weighted average from regime anchor.
- **Breadth and flow**:
  - McClellanOsc: breadth-style oscillator.
  - WeightedAlpha: time-weighted return strength.
  - WilderAccSwingIndex: cumulative directional bias.
  - AccDistWill, AccDistWillMovAvg: accumulation/distribution flow.
- **Volatility regime**:
  - IV_High, IV_Mid, IV_Low: implied volatility bands.
- **Options chain aggregates**:
  - Options_Total_Volume, Options_Put_Volume, Options_Call_Volume, OSC_Volume.
- **Microstructure**:
  - aggregate_bid_count, aggregate_ask_count, mean_bid, mean_ask, quote_spread.

Feature Families and Ablation Experiments

Feature Families

1. **Price Dynamics**: open, high, low, close, log_return, ret_z, atr_pct.
2. **Volatility Structure**: vol_ewma, kappa_proxy, vol_energy, IV bands.
3. **Trend and Momentum**: sma, FRAMA, Anchored VWAP, MACD, RSI, ADX.
4. **Bands and Consolidation**: Bollinger bands, bandwidth, breakout_score.
5. **Volume and Flow**: cmf, AccDistWill, OSC_Volume, pressure metrics.
6. **Breadth and Regime**: McClellanOsc, WeightedAlpha, chaos_membership.
7. **Options Surface**: Greeks, IV, IVR, TE.
8. **Microstructure**: bid/ask counts, spread, friction.
9. **Fuzzy Logic and Control**: fuzzy_reversion, exec_allow, risk_override.

Ablation Experiments

- **Ablate microstructure**: Remove aggregate_bid_count, mean_bid, quote_spread → test impact on execution-aware modeling.
- **Ablate IV regime**: Remove IV_High/Mid/Low → test volatility positioning sensitivity.
- **Ablate breadth**: Remove McClellanOsc, WeightedAlpha → test regime classification degradation.
- **Ablate fuzzy logic**: Remove chaos_membership, fuzzy_reversion → test gating and predicate stability.
- **Ablate chain aggregates**: Remove Options_Total_Volume, Put/Call volumes → test sentiment signal loss.

Each experiment measures impact on CondorNet's composite loss function (NPDD, Sharpe, drawdown, fuzzy loss, entropy, etc.).

Reversal-Specific Composite Features

1. Reversal_Alpha_1

$$Reversal\_Alpha\_1 = (FRAMA < Close)\ AND\ (AnchoredVWAP < Close)\ AND\ (IV\_Mid < IV\_Low + 0.2 * (IV\_High - IV\_Low))\ AND\ (McClellanOsc > McClellanOsc[5])$$

Interpretation: Price above trend baselines, IV compression, breadth divergence → bottom reversal.

2. Reversal_Alpha_2

$$Reversal\_Alpha\_2 = (Close < FRAMA)\ AND\ (Close < AnchoredVWAP)\ AND\ (IV\_Mid > IV\_High - 0.2 * (IV\_High - IV\_Low))\ AND\ (McClellanOsc < McClellanOsc[5])$$

Interpretation: Price below trend, IV exhaustion, breadth breakdown → top reversal.

3. Reversal_Score_Composite

$$Reversal\_Score = w1 * BB\_Percentile + w2 * RSI\_Dyn + w3 * McClellanOsc + w4 * IV\_Position + w5 * AccDistWillMovAvg$$

Where $IV\_Position = normalized\ IV\_Mid$ within band. Weights tuned via feature dominance analysis.

These composite features are used as inputs to CondorNet's fuzzy gating and regime classification blocks.

This framework supports robust experimentation, interpretability, and model evolution for CondorNet v4.0 → CondorNet v5.0

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